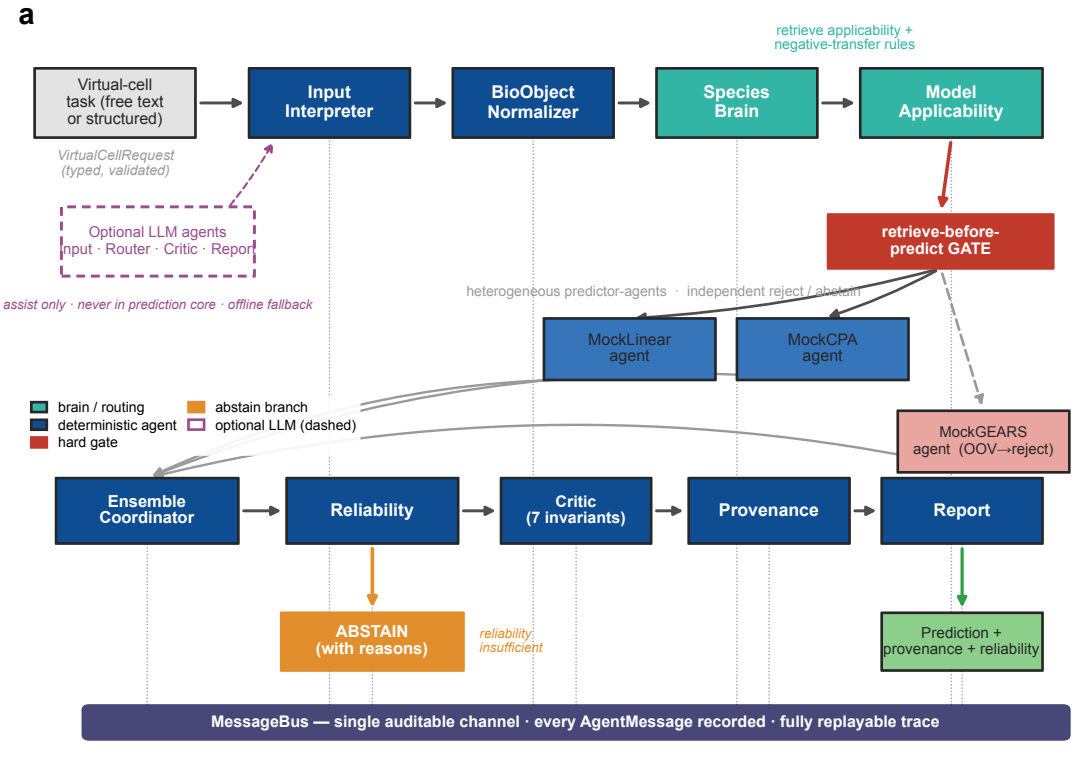

a
retrieve applicability +
negative-transfer rules
Virtual-cell
task (free text
or structured)
Input
Interpreter
BioObject
Normalizer
Species
Brain
Model
Applicability
VirtualCellRequest
(typed, validated)
Optional LLM agents
nput · Router · Critic · Repor
retrieve-before-
predict GATE
assist only · never in prediction core · offline fallback
heterogeneous predictor-agents · independent reject / abstain
MockLinear
agent
MockCPA
agent
brain / routing
deterministic agent
hard gate
abstain branch
optional LLM (dashed)
MockGEARS
agent (OOV→reject)
Ensemble
Coordinator
Reliability
Critic
(7 invariants)
Provenance
Report
ABSTAIN
(with reasons)
reliability
insufficient
Prediction +
provenance + reliability
MessageBus — single auditable channel · every AgentMessage recorded · fully replayable trace